

PRATEEK SINGH

Computer Vision Engineer | AI / ML Engineer | Edge AI & GenAI Specialist

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PROFESSIONAL SUMMARY

Computer Vision and AI/ML Engineer with 2+ years of production experience building deep learning pipelines, real-time video analytics systems, and edge AI deployments from prototype to scale. Shipped **YOLO-based multi-modal detection** achieving **97% mAP** on live retail streams running on **Hailo-8** and **NVIDIA GPU** hardware across 7+ international stores. Deep expertise across the full CV stack — **GStreamer, RTSP, Re-ID, privacy-preserving face blur, trigger-based tracking, LOR/ROI video filtering**, and **Kafka**-driven inference architectures. Extended into **Generative AI** (LangChain, RAG, Whisper, GPT-4o-mini) and full-stack **AI SaaS** (Next.js, TypeScript, Prisma). Independently shipped **6+ AI/CV projects** spanning security analytics, crowd intelligence, face anonymisation, and SaaS platforms. Available immediately.

CORE COMPETENCIES

Real-Time Video Analytics · Object Detection & Tracking · Edge AI Deployment · Privacy-Preserving CV · Multi-Camera RTSP Pipelines · Re-ID & Person Tracking · LOR / ROI Event Filtering · MLOps & Model Deployment · RAG / LLM Integration · Multi-Agent AI Systems · AI SaaS Product Development · School & Public Safety CV

TECHNICAL SKILLS

CV & Detection	YOLO v5/v7/v8/v8n-face, Object Detection, Image Classification, Semantic Segmentation, Pose Estimation, Re-ID (torchreid), Multi-modal Detection, Video Analytics, Crowd Analytics, Face Anonymisation, LOR/ROI Filtering, Privacy-preserving CV
Deep Learning	PyTorch, TensorFlow, Keras, Darknet, Ultralytics, CNN, Transfer Learning, Fine-tuning, PIFuHD 3D Reconstruction, Feature Extraction
Edge AI & Infra	Hailo-8, NVIDIA GPU, CUDA, cuDNN, Embedded C++, GStreamer, RTSP/NVR Streaming, On-device Inference, Batch Video Processing, Socket Programming, Low-latency Systems, Raspberry Pi
GenAI & LLMs	LangChain, RAG, Vector Databases, Embeddings, HuggingFace, Whisper, GPT-4o-mini, Transformer Models, Multi-agent Systems
Full-Stack AI	Next.js, TypeScript, React, Prisma, Node.js, Clerk Auth, Telegram Bot API, Notion API, Google Drive API, SerpAPI, Hunter.io, Resend
MLOps & Cloud	Docker, CI/CD, MLOps, AWS, GCP, Azure, Linux, Shell Scripting, GitHub Actions, Model Zoo, Model Deployment
Data & APIs	Kafka, REST APIs, MongoDB, SQL, OpenCV, NumPy, Pandas, Postman, Redash, Data Annotation, Label Transformation

PROFESSIONAL EXPERIENCE

Computer Vision Engineer — Service Automation *Autonomo Technologies* Aug 2024 – Apr 2026

Bengaluru, India (Hybrid)

- Optimised and deployed** YOLOv7 deep learning models for person detection, hand-pose estimation, and multi-modal image classification across live **German retail stores**; achieved **97% mAP** on proprietary datasets with real-time inference on **Hailo-8** and **NVIDIA GPU** edge hardware, scaled to **7+ stores** with zero downtime over 6+ months.
- Engineered** end-to-end **automated ML pipelines** (data processing, annotation, validation, label transformation, training workflows), cutting manual effort by **70%** and establishing a reusable in-house **model zoo** for standardised multi-store deployment.
- Integrated** CV with IoT infrastructure via **Kafka** message queues, **REST APIs**, and distributed payload architectures, enabling scalable real-time ML detection pipelines across production retail environments.
- Designed** ML-based **Re-ID** and neural network pipelines with multi-source recommendation systems, improving person-tracking continuity and operational efficiency across distributed retail locations.
- Built** GenAI and LLM integration experiments using **LangChain** and **HuggingFace**, applying **RAG**-based retrieval with vector embeddings for CV pipeline automation and intelligent service decision-making.

Computer Vision Engineer Intern — AI/ML & Edge AI *Autonomo Technologies* Feb 2024 – Aug 2024

Bengaluru, India (Hybrid)

- Developed** deep learning models for person detection, hand-object identification, and pose estimation on **Darknet**; achieved **90%+ mAP** through iterative transfer learning and fine-tuning on custom retail datasets.
- Deployed** low-latency neural networks with **CUDA/cuDNN**; built **Embedded C++** pipelines for object detection, Re-ID, and **RTSP stream processing** via **GStreamer** for high-performance real-time CV in production.
- Automated** data analysis and API workflows via **Postman** and **Redash**, streamlining ML model validation and re-

porting across CV pipeline components.

- **Implemented** autonomous CV systems including **fire safety triggers** and depth optimisation, expanding platform capabilities into real-world safety and environmental monitoring.

PROJECTS — RANKED BY TECHNICAL IMPACT

1. AI-BlockFilter-By-LOR — Security Video Segment Intelligence [GH:repo]

Python • YOLO • RTSP/NVR • Line/Region-of-Interest • OpenCV • JSON Config

- **Architected** a production-ready pipeline converting continuous **NVR/RTSP streams** into meaningful event blocks by running YOLO person detection against user-defined **Lines/Regions-of-Interest (LOR)**; auto-extracts, compresses, labels, and stores only segments where persons cross LOR boundaries.
- Eliminates false-alarm noise for security operators; architecture directly maps to **school security, campus access control**, and multi-site surveillance with zero changes to existing camera infrastructure.

2. Trigger_Based_Tracking — Region-Triggered Person Re-ID Pipeline [GH:repo]

Python • torchreid • Re-ID • ROI Pairing • Entry/Exit Detection • OpenCV

- **Built** a trigger-driven tracking system: on any region/event trigger, **torchreid Re-ID** activates to follow individuals across camera views within a premises using **ROI-to-person pairing** and entry/exit activity detection.
- Production-structured with config-driven components (person extraction, product-person pairing, activity handler) — ready to integrate into multi-camera **CCTV security** and retail loss-prevention systems.

3. Custom-Face-Blur-PS — Privacy-Preserving Face Anonymisation Pipeline [GH:repo]

Python • YOLOv8n-face • OpenCV • Elliptical Blur • GPU • Batch Video • Shell

- **Developed** a production-ready batch script applying **oval/elliptical face blurring** across entire video folders using **YOLOv8n-face** detection; GPU-accelerated, preserves original resolution, codec, and FPS.
- Deployable directly in **GDPR-sensitive CCTV, school safeguarding**, and public-space privacy contexts via shell launcher with argument parsing and auto-model download — no hardware changes required.

4. CV-SPINE / VibeCheck — Real-Time Crowd Analytics & Safety System [GH:repo]

Python • CV Pipeline • GStreamer • RTSP • Rolling Window Aggregation • Dashboard

- **Designed** a privacy-first CV pipeline ingesting live camera streams, extracting crowd signals (density, movement, pose, attention) over **rolling time windows**, and generating real-time human-readable crowd-state descriptors.
- Five-function signal architecture (F1-F5) mapping raw inference to actionable crowd intelligence; live **web dashboard** with rule-based engine — directly applicable to **school safety and venue crowd monitoring**.

5. Client-Anchor — AI-Powered Lead Discovery & Outreach SaaS [GH:repo]

Next.js • TypeScript • Prisma • Clerk • SerpAPI • Hunter.io • Resend • XLSX

- **Shipped** full-stack **AI SaaS**: natural-language search → multi-source aggregation (Google Places, SerpAPI, Adzuna) → 3-tier contact enrichment → automated cold outreach via Resend; 17-column XLSX/CSV export, background search refresh, Clerk auth.
- Demonstrates end-to-end **LLM product architecture**: intent inference, multi-API orchestration, and personalised outreach generation at production scale in Next.js.

6. 3D Object Reconstruction — Deep Learning Spatial CV Research [GH:repo]

Python • PIFuHD • Deep Learning • 3D Reconstruction Metrics

- **Implemented** and evaluated **PIFuHD** (Pixel-Aligned Implicit Function for High-Resolution 3D Human Digitisation); assessed reconstruction accuracy metrics and improvement strategies — extends CV expertise into spatial and volumetric modelling domains.

KEY ACHIEVEMENTS

- **97% mAP** with YOLOv7 on custom multi-class retail dataset — highest accuracy on internal benchmark across all deployed models.
- **7+ live retail stores in Germany** running real-time CV inference with zero downtime over 6+ months of continuous production deployment.
- **70% reduction** in ML pipeline deployment time by building an in-house automated training, validation, and model zoo workflow.
- **Multi-modal detection** (person + hand-pose + object) running concurrently on a single Hailo-8 edge device within real-time latency budget — no cloud dependency.
- Independently shipped **6+ AI/CV projects** across security analytics, privacy-preserving CV, crowd intelligence, and GenAI SaaS products.
- Built **GDPR-ready face anonymisation** pipeline deployable on existing camera infrastructure with zero hardware changes.

EDUCATION

B.Tech — Computer Science & Engineering (AI/ML Specialisation)

2020 – 2024

BML Munjal University, Gurugram, Haryana CGPA: **8.0 / 10**

Open to: Computer Vision Engineer • AI/ML Engineer • Deep Learning Engineer • GenAI/LLM Engineer • MLOps Engineer • Noida / Gurugram / NCR / Remote

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